(iii) If $p > 1$ and p' satisfies $\frac{1}{p} + \frac{1}{p'} = 1$, then the dual of $\mathcal{T}_p^{(\alpha)}(t^\alpha)$ is $\mathcal{T}_{p'}^{(\alpha)}(t^\alpha)$, where the duality is given by

$$\langle f, g\rangle_\alpha = \frac{1}{\Gamma(\alpha+1)^2}\int_0^\infty W_+^\alpha f(t) W_+^\alpha g(t) t^{2\alpha} dt = \langle D_+^\alpha f, D_+^\alpha g\rangle_0,$$

for $f \in \mathcal{T}_p^{(\alpha)}(t^\alpha)$, $g \in \mathcal{T}_{p'}^{(\alpha)}(t^\alpha)$.

If $\alpha, a > 0$ and $p \geq 1$, then
(i) $t^\beta \notin \mathcal{T}_p^{(\alpha)}(t^\alpha)$ for $\beta \in \mathbb{C}$.
(ii) $(a+t)^{-\beta} \in \mathcal{T}_p^{(\alpha)}(t^\alpha)$ for $\Re\beta > 1/p$. ([25, Lemma 2.3]).

Remark 3.1 For $p = \infty$, the nature of the space $\mathcal{T}_\infty^{(\alpha)}(t^\alpha)$ is different than $\mathcal{T}_p^{(\alpha)}(t^\alpha)$ for $p \geq 1$. One might think to define

$$||f||_{\alpha,\infty} := \frac{1}{\Gamma(\alpha+1)}\text{ess sup}\{t^\alpha |W_+^\alpha f(t)| \ : \ t \in \mathbb{R}^+\} = ||D_+^\alpha f||_\infty < \infty.$$

Consider $l(t) := \log(1+t)$ $(t \in \mathbb{R}^+)$; note that $||l||_{1,\infty} < \infty$ and $\|l\|_\infty = +\infty$. However in the case that $0 < \alpha < \beta$, note that for $t > 0$,

$$|D_+^\alpha f(t)| \leq \frac{t^\alpha}{\Gamma(\alpha+1)}\int_t^\infty \frac{(s-t)^{\beta-\alpha-1}}{s^\beta \Gamma(\beta-\alpha)} s^\beta |W_+^\beta f(s)| ds \leq \frac{\beta}{\alpha}||f||_{\beta,\infty},$$

and we conclude $||f||_{\alpha,\infty} \leq \frac{\beta}{\alpha}||f||_{\beta,\infty}$ for f in the Schwarz class on $\mathbb{R}^+$. Due to, we will skip the case $p = \infty$ on this paper, although some of next results are also valid in this particular case.

In the next theorem, we extend the Hölder inequality in Lebesgue spaces $L^p(\mathbb{R}^+)$ to the case of fractional Sobolev-Lebesgue spaces $\mathcal{T}_p^{(\alpha)}(t^\alpha)$. Note that in the integer case $\alpha \in \mathbb{N}$, it a straightforward consequence of Leibnitz formula. To attack the general case we use the following Leibnitz formula

$$W_+^\alpha(fg)(x) = f(x)W_+^\alpha g(x) + g(x) W_+^\alpha f(x) - \int_x^\infty \int_x^\infty \frac{d\varphi_{t,u}^{\alpha-1}(x)}{dx} W_+^\alpha f(t) W_+^\alpha g(u) dt du$$

for $f, g \in \mathcal{S}_+$ and $\alpha > 0$ [17, Proposition 2.5], where the function $\varphi_{t,u}^{\alpha-1}$ is given by

$$\varphi_{t,u}^{\alpha-1}(x) = \frac{(u-x)^{\alpha-1}}{\Gamma(\alpha)}\, {}_2F_1\left(-\alpha+1, -\alpha+1, 1; \frac{t-x}{u-x}\right),$$

for $x < t < u$; $\varphi_{t,u}^{\alpha-1}(x) = \varphi_{u,t}^{\alpha-1}(x)$ for $x < u < t$ and let $\varphi_{t,u}^{\alpha-1} = 0$ in all other cases ([17, p. 313]).